

Youssef Boughanmi

Machine Learning & Software Developer

✉ youssefboughanmy@gmail.com ☎ +216 26 068 101

🌐 linkedin.com/in/youssef-boughanmi-4990222a0 🐙 github.com/boughanmiyoussef

🔗 boughanmiyoussef.github.io

SUMMARY

Machine Learning and Software Developer with a portfolio of applied AI systems spanning Retrieval-Augmented Generation (RAG), computer vision, time-series forecasting, and data engineering pipelines. Skilled in Python, SQL, Scikit-Learn, and cloud-ready deployment using Docker and Streamlit. Self-directed builder who ships working, deployed systems end-to-end, from data ingestion to production inference, with measurable results across every project.

SKILLS

Programming Languages:

Python, SQL, T-SQL, JavaScript (ES6), C

Machine Learning & AI:

Machine Learning, Scikit-Learn, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), LangChain, Natural Language Processing (NLP), Time Series Forecasting, Hyperparameter Optimization (Optuna), Model Interpretability (SHAP), Computer Vision (OpenCV)

Data Engineering:

ETL Pipelines, Data Warehousing, Medallion Architecture (Bronze/Silver/Gold), Star Schema Design, Data Pipeline Optimization, Pandas, NumPy

Cloud, DevOps & Deployment:

Docker, Git, Linux, REST APIs, Streamlit, Render, Cloud Deployment

Data Visualization & Analytics:

Plotly, Statistical Analysis, Anomaly Detection

PROJECTS

FastYolo — Real-Time Object Detection on Edge Devices 🔗

08/2026 – 09/2026

Technologies: Python, OpenCV, YOLOv4-Tiny, Computer Vision, Docker

- Implemented a CPU-only real-time object detection pipeline using YOLOv4-Tiny, optimized for deployment on resource-constrained edge hardware such as Raspberry Pi.
- Engineered an edge-profile simulator to benchmark model performance under constrained hardware envelopes (limited CPU threads, reduced input resolution) without requiring physical edge devices.
- Built a full evaluation suite including FPS/latency benchmarking and mAP/precision/recall assessment on COCO validation data.
- Containerized the application with Docker to replicate a Raspberry Pi 4B hardware profile, ensuring reproducible cross-platform inference via OpenCV's DNN module.

DocuMint - Intelligent Document Q&A System 🔗

06/2026 – 07/2026

Technologies: Python, LangChain, ChromaDB, Ollama, Streamlit, RAG, NLP

- Architected an end-to-end Retrieval-Augmented Generation (RAG) pipeline for automated document ingestion, processing over 1,500 text chunks for semantic search.
- Implemented high-performance vector similarity search using ChromaDB, integrated with multi-model Large Language Model (LLM) deployment via Ollama (phi3:mini, tinyllama).
- Engineered a responsive Streamlit web application featuring drag-and-drop PDF upload functionality, optimized for low-resource environments.

- Achieved sub-5-second query response times on 8GB RAM hardware, demonstrating efficient local LLM inference optimization.

StockSentinel - Financial Anomaly Detection System

03/2026 – 04/2026

Technologies: Python, Scikit-Learn, Pandas, Plotly, Streamlit, APIs, Render

- Developed a real-time anomaly detection system to flag financial market volatility spikes using unsupervised machine learning.
- Implemented Isolation Forest and Z-score statistical methods for robust unsupervised outlier detection.
- Engineered streaming financial indicators, including RSI, volatility metrics, and volume-based signals, for live market monitoring.
- Optimized the data ingestion pipeline to reduce latency by 40%, enabling real-time volatility monitoring through an interactive dashboard.

BikeDemand – Temporal Demand Forecasting Platform

01/2026 – 02/2026

Technologies: Python, Scikit-Learn, Optuna, SHAP, Time Series Analysis

- Designed a machine learning forecasting system to predict hourly bike-sharing demand using historical usage data.
- Implemented a time-series machine learning pipeline with engineered cyclical temporal features to capture seasonality patterns.
- Applied TimeSeriesSplit cross-validation and automated hyperparameter optimization (Optuna) to ensure model robustness and generalization.
- Achieved an R^2 of 0.9162 and RMSE of 63.62 on validation data, demonstrating high predictive accuracy.

FactCheck - AI Fake News Detection System

06/2025 – 07/2025

Technologies: Python, Scikit-learn, Pandas, NLTK, TF-IDF, Logistic Regression

- Developed a Natural Language Processing (NLP) classification system to detect fake news across large-scale text datasets.
- Implemented TF-IDF vectorization combined with Logistic Regression for efficient and interpretable text classification.
- Applied k-fold cross-validation to ensure model generalization, stability, and resistance to overfitting.
- Achieved 95.2% classification accuracy and a 0.990 ROC-AUC score across 20,000+ news articles.

MedallionDB – Data Warehouse & ETL Pipeline

02/2025 – 03/2025

Technologies: SQL Server, T-SQL, Docker, Medallion Architecture, Data Warehousing

- Architected a Medallion-style data warehouse with Bronze, Silver, and Gold layers to support scalable analytics.
- Implemented ETL workflows using SQL Server stored procedures and custom transformation logic.
- Modeled a star schema optimized for analytical query performance and downstream reporting.
- Containerized the full data pipeline using Docker to ensure a reproducible, portable data engineering environment.

EDUCATION

Full-Stack Software Developer – Machine Learning Specialization

12/2024 – 05/2026

Holberton School 

Tunis, Tunisia

High School Diploma

09/2018 – 06/2022

Taha Hussein 

Mégrine

LANGUAGES

English



French

